

Lecture 11

Boundary-Value Problems
Finite Difference Method
system assembly with Dirichlet and Robin BC's
convection-diffusion problem and upwinding

Boundary-Value Problems in One Dimension

BVP for Poisson equation with Dirichlet BC's

This the 1D Boundary-Value Problem (BVP) for the Poisson equation with Dirichlet Boundary Conditions (BC's) and three different source functions:

$$-u'' = f, \quad x \in (0, 1)$$

$$u(0) = 0, \quad u(1) = 0$$

case 1: $f(x) = -1$

case 2: $f(x) = -x + x^2 - x^3 + x^4 - x^5 + x^6$

case 3: $f(x) = -\sin(\pi x)$

Analytical solutions

This is an ordinary linear second-order differential equation, which can be solved analytically:

case 1: $f(x) = -1$;

$$u = -\frac{1}{2}x + \frac{1}{2}x^2$$

case 2: $f(x) = -x + x^2 - x^3 + x^4 - x^5 + x^6$;

$$u = -\frac{1}{840}(15x^8 - 20x^7 + 28x^6 - 42x^5 + 70x^4 - 140x^3 + 89x)$$

case 3: $f(x) = -\sin(\pi x)$;

$$u = -\frac{1}{\pi^2} \sin(\pi x)$$

Types of Boundary Conditions

- Dirichlet BC: $u(a) = g_a$
- Neumann BC: $u'(a) = v_a$
- Robin (mixed) BC: $-u'(a) + \beta u(a) = \mu_a$

BVP for Poisson equation with mixed BC's

$$\begin{aligned} -u'' &= f, \quad x \in (a, b) \\ -u'(a) + \beta u(b) &= \mu_a, \quad u(b) = \mu_b \end{aligned}$$

This problem describes the stationary distribution of temperature in a rod with one end ($x = b$) kept at temperature μ_b , while the other end ($x = a$) is exchanging heat with environment ($\beta \geq 0$).

BVP for convection-diffusion equation

Steady-state convection-diffusion equation with Dirichlet BC's:

$$\begin{aligned} -\varepsilon u'' + vu' &= 0, \\ u(0) &= 0, \quad u(1) = 1 \end{aligned}$$

$u(x)$ – concentration of substance (e.g. pollutant)

ε – diffusion constant

v – velocity of the convecting medium

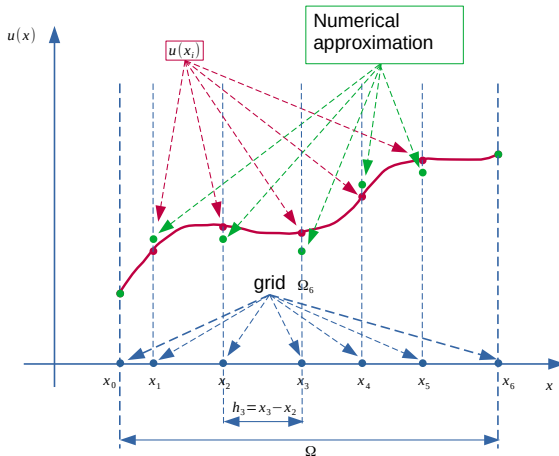
Analytical solution:

$$u(x) = \frac{e^{\frac{vx}{\varepsilon}} - 1}{e^{\frac{v}{\varepsilon}} - 1}$$

Péclet number, $Pe = vL/\varepsilon$, where L is the length of the domain, determines the relative "magnitude" of convection with respect to diffusion.

Finite-Difference Method (FDM)

Idea of numerical solution



Steps of FDM

$$\begin{aligned} -u'' &= f, \quad x \in (a, b) \\ u(a) &= g_a, \quad u(b) = g_b \end{aligned}$$

- Introduce a grid of $N + 1$ points x_i , $i = 0, \dots, N$, with $x_0 = a$, $x_N = b$.
- Use FD approximation of $-u''(x_i) = -u''_i$ at inner points x_i , $i = 1, \dots, N - 1$
- Use BC's at the boundary points x_0 and x_N
- Assemble linear system and solve it

Approximating differential equation at inner points

Let $u(x)$ be a sufficiently smooth function defined on (a, b) . Introducing notation: $f_i = f(x_i)$ and $u_i = u(x_i)$, we write the differential equation at inner points:

$$-u_i'' = f_i, \quad i = 1, 2, \dots, N-1,$$

Using a Finite-Difference formula for $-u_i''$

In Chapter on Numerical Differentiation (Lecture 4) we have derived:

$$u''(x) = \frac{u(x-h) - 2u(x) + u(x+h)}{h^2} - \frac{u^{(4)}(\xi)}{6}h^2,$$

Hence

$$-u_i'' = \frac{-u_{i+1} + 2u_i - u_{i-1}}{h^2} + \mathcal{O}(h^2)$$

FD method in a nutshell

Continuous problem:

$$-\frac{\partial^2 u}{\partial x^2} = f, \quad x \in (0, 1)$$

$$u(0) = 0, \quad u(1) = 0$$

case 1: $f(x) = -1$

case 2: $f(x) = -x + x^2 - x^3 + x^4 - x^5 + x^6$

case 3: $f(x) = -\sin(\pi x_i)$

FD method: choose $h = 1/N$, define $x_i = 0 + hi$, $i = 0, \dots, N$, and solve

$$\frac{-u_{i+1} + 2u_i - u_{i-1}}{h^2} = f_i, \quad i = 1, \dots, N-1$$

$$u_0 = 0, \quad u_N = 0$$

case 1: $f_i = -1$

case 2: $f_i = -x_i + x_i^2 - x_i^3 + x_i^4 - x_i^5 + x_i^6$

case 3: $f_i = -\sin(\pi x_i)$

Discrete equations of FDM

Discrete FD equations:

$$u_0 = g_a \quad \text{left BC}$$

$$\frac{1}{h^2}(-u_0 + 2u_1 - u_2) = f_1$$

$$\frac{1}{h^2}(-u_1 + 2u_2 - u_3) = f_2$$

...

$$\frac{1}{h^2}(-u_{N-2} + 2u_{N-1} - u_N) = f_{N-1}$$

$$u_N = g_b \quad \text{right BC}$$

Elimination of BC's

Boundary values $u_0 = g_a$ and $u_N = g_b$ are eliminated and the BC's are shifted to the right-hand side:

$$\frac{1}{h^2}(2u_1 - u_2) = f_1 + \frac{1}{h^2}g_a$$

$$\frac{1}{h^2}(-u_1 + 2u_2 - u_3) = f_2$$

...

$$\frac{1}{h^2}(-u_{N-2} + 2u_{N-1}) = f_{N-1} + \frac{1}{h^2}g_b$$

System Assembly

$$\frac{1}{h^2} \begin{bmatrix} 2 & -1 & 0 & 0 & \dots & 0 \\ -1 & 2 & -1 & 0 & \dots & 0 \\ \cdot & & & & & \cdot \\ \cdot & & & & & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ \cdot \\ \cdot \\ u_{N-1} \end{bmatrix} = \begin{bmatrix} f_1 + g_a/h^2 \\ f_2 \\ \cdot \\ \cdot \\ f_{N-1} + g_b/h^2 \end{bmatrix}$$

In short linear-algebra notation:

$$\mathbf{L}\mathbf{u} = \mathbf{f}$$

The system matrix $\mathbf{L} \in \mathbb{R}^{(N-1) \times (N-1)}$ is an $\mathcal{O}(h^2)$ FD approximation of the negative one-dimensional Laplacian with Dirichlet BC's at both ends of the interval.

Solving linear algebraic problem

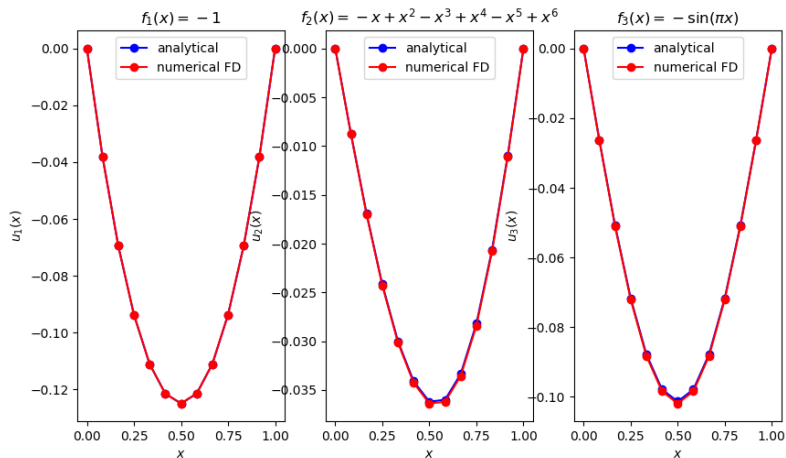
Assuming that the system matrix L is invertible

$$L\mathbf{u} = \mathbf{f}$$

$$\mathbf{u} = L^{-1}\mathbf{f}$$

The resulting vector $\mathbf{u} \in \mathbb{R}^{N-1}$ contains the FDM approximation of the grid-values of the solution $u(x_i)$ at the inner points of the grid $i = 1, \dots, N - 1$.

Matplotlib figures



FDM with mixed BC's

BVP for Poisson equation with mixed BC's

$$\begin{aligned} -u'' &= f, \quad x \in (a, b) \\ -u'(a) + \beta u(a) &= \mu_a \\ u(b) &= \mu_b \end{aligned}$$

We consider uniform grid: $x_i = a + hi$, $i = 0, \dots, N$, $h = (b - a)/N$.

FD equations at inner points

Using $\mathcal{O}(h^2)$ FD approximation for $-u_i''$ and eliminating Dirichlet BC, we write down:

$$\begin{aligned}\frac{1}{h^2}(u_0 - 2u_1 - u_2) &= f_1, \\ &\vdots \\ \frac{1}{h^2}(-u_{N-2} + 2u_{N-1}) &= f_{N-1} + \frac{1}{h^2}\mu_b\end{aligned}$$

However, Robin BC:

$$-u_0' + \beta u_0 = \mu_a,$$

Contains the derivative u_0' , which requires FD approximation.

Robin BC



Using the forward one-sided FD approximation $u'_0 = (u_1 - u_0)/h + \mathcal{O}(h)$, we complete the system as:

$$\frac{1}{h}(u_0 - u_1) + \beta u_0 = \mu_a$$

$$\frac{1}{h^2}(u_0 - 2u_1 - u_2) = f_1,$$

$$\vdots$$

$$\frac{1}{h^2}(-u_{N-2} + 2u_{N-1}) = f_{N-1} + \frac{1}{h^2}\mu_b$$

Symmetrization

$$\frac{1}{h^2} ((1 + h\beta)u_0 - u_1) = \frac{\mu_a}{h}$$

$$\frac{1}{h^2}(u_0 - 2u_1 - u_2) = f_1,$$

$$\vdots$$

$$\frac{1}{h^2}(-u_{N-2} + 2u_{N-1}) = f_{N-1} + \frac{\mu_b}{h^2}$$

with this changes in the first FD equation, the system matrix will be symmetric.

System assembly

$$\frac{1}{h^2} \begin{bmatrix} (1+h\beta) & -1 & 0 & 0 & \dots & 0 \\ -1 & 2 & -1 & 0 & \dots & 0 \\ \cdot & & & & & \cdot \\ \cdot & & & & & -1 \\ 0 & 0 & 0 & 0 & -1 & 2 \end{bmatrix} \begin{bmatrix} u_0 \\ u_1 \\ \cdot \\ \cdot \\ u_{N-1} \end{bmatrix} = \begin{bmatrix} \mu_a/h \\ f_1 \\ \cdot \\ \cdot \\ f_{N-1} + \mu_b/h^2 \end{bmatrix}$$

It still has the form $L\mathbf{u} = \mathbf{f}$, but

- $L \in \mathbb{R}^{N \times N}$ is the matrix of the negative FD Laplacian with a Robin BC on one side and a Dirichlet BC at the other end of the interval
- $\mathbf{u} \in \mathbb{R}^N$ contains one additional entry, u_0
- $\mathbf{f} \in \mathbb{R}^N$ is different

FDM with upwinding for convection-diffusion BVP

Convection-diffusion BVP

$$\begin{aligned} -\varepsilon c'' + vc' &= 0, \\ c(0) &= 0, \quad c(1) = 1 \end{aligned}$$

$c(x)$ – concentration of substance (e.g. pollutant)

ε – diffusion constant

v – velocity of the convecting medium

Analytical solution:

$$c(x) = \frac{e^{\frac{vx}{\varepsilon}} - 1}{e^{\frac{v}{\varepsilon}} - 1}$$

Péclet number, $Pe = vL/\varepsilon$, where L is the length of the domain, determines the relative "magnitude" of convection with respect to diffusion.

Using $\mathcal{O}(h^2)$ FD approximations, inner points

$$\begin{aligned} -\varepsilon c_i'' + v c_i' &\approx \varepsilon \frac{-c_{i-1} + 2c_i - c_{i+1}}{h^2} + v \frac{c_{i+1} - c_{i-1}}{2h} \\ &= \varepsilon \left(\frac{-c_{i-1} + 2c_i - c_{i+1}}{h^2} + \frac{vh}{2\varepsilon} \frac{c_{i+1} - c_{i-1}}{h^2} \right) = 0 \end{aligned}$$

where mesh Péclet number is defined as

$$p_h = \frac{vh}{2\varepsilon} = \frac{h}{2L} \text{Pe} = \frac{1}{2N} \text{Pe}$$

FD equations

$$\begin{aligned} (-1 - p_h)c_{i-1} + 2c_i + (-1 + p_h)c_{i+1} &= 0 \\ i &= 2, 3, \dots, N-2 \end{aligned}$$

Eliminating Dirichlet BC's

Left boundary

$$-\varepsilon c_1'' + v c_1' = 0, \text{ and } c_0 = 0$$

$$\frac{\varepsilon}{h^2} [(-1 - p_h)c_0 + 2c_1 + (-1 + p_h)c_2] = 0$$

$$2c_1 + (-1 + p_h)c_2 = 0$$

Right boundary

$$-\varepsilon c_{N-1}'' + v c_{N-1}' = 0, \text{ and } c_N = 1$$

$$\frac{\varepsilon}{h^2} [(-1 - p_h)c_{N-2} + 2c_{N-1} + (-1 + p_h)c_N]$$

$$= \frac{\varepsilon}{h^2} ((-1 - p_h)c_{N-2} + 2c_{N-1} + (-1 + p_h)1) = 0$$

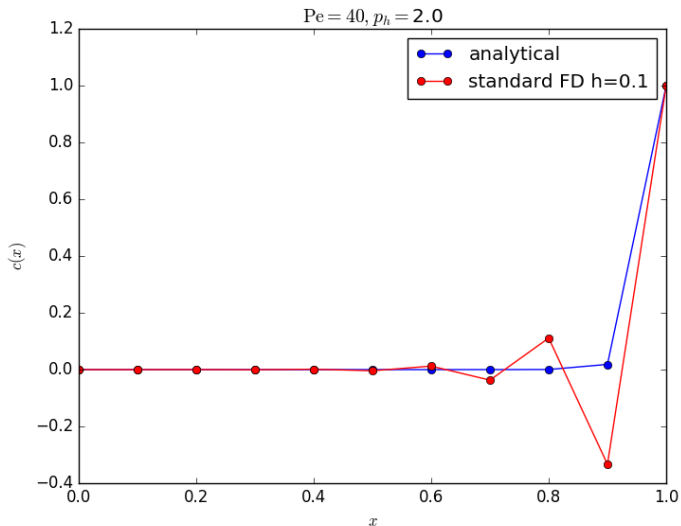
$$(-1 - p_h)c_{N-2} + 2c_{N-1} = 1 - p_h$$

System assembly

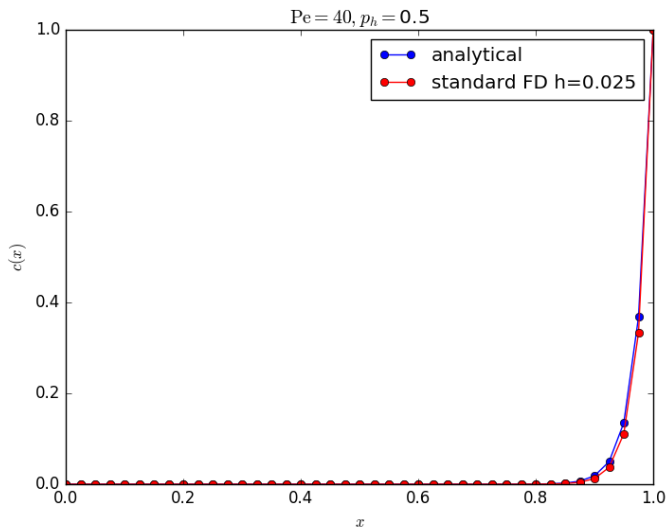
$$A\mathbf{c} = \mathbf{f}$$

$$\begin{bmatrix} 2 & -1 + p_h & 0 & 0 & \dots & 0 \\ -1 - p_h & 2 & -1 + p_h & 0 & \dots & 0 \\ \cdot & & & & & \cdot \\ \cdot & & & & & -1 + p_h \\ 0 & 0 & 0 & 0 & -1 - p_h & 2 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_{N-1} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ 1 - p_h \end{bmatrix}$$

Results with $h = 0.1$



Results with $h = 0.025$



Why?

Analytical solution of FD equation:

$$(-1 - p_h)c_{i-1} + 2c_i + (-1 + p_h)c_{i+1} = 0, \quad c_0 = 0, \quad c_N = 1$$

take: $c_i = r^i$

$$(-1 - p_h)r^{i-1} + 2r^i + (-1 + p_h)r^{i+1} = 0$$

$$-1 - p_h + 2r + (-1 + p_h)r^2 = 0$$

$$r = 1 \quad \text{and} \quad r = \frac{-1 - p_h}{-1 + p_h} = \frac{1 + p_h}{1 - p_h}$$

$$c_i = A + Br^i = A + B \left(\frac{1 + p_h}{1 - p_h} \right)^i$$

$$c_i = \frac{\left(\frac{1 + p_h}{1 - p_h} \right)^i - 1}{\left(\frac{1 + p_h}{1 - p_h} \right)^N - 1}, \quad \text{monotonically increasing if } \frac{1 + p_h}{1 - p_h} \geq 0$$

The mesh Péclet number must be $p_h = \nu h / (2\varepsilon) = \text{Pe } h / (2L) < 1$. Hence, the grid step must be $h < 2L / \text{Pe}$

The unwinding trick

The idea is to use the backward FD approximation for vc' if $v > 0$, and the forward FD approximation if $v < 0$. For example ($v > 0$):

$$\begin{aligned} -\varepsilon c_i'' + vc_i' &\approx \varepsilon \frac{-c_{i-1} + 2c_i - c_{i+1}}{h^2} + v \frac{c_i - c_{i-1}}{h} = 0 \\ (-1 - 2p_h)c_{i-1} + (2 + 2p_h)c_i - c_{i+1} &= 0 \end{aligned}$$

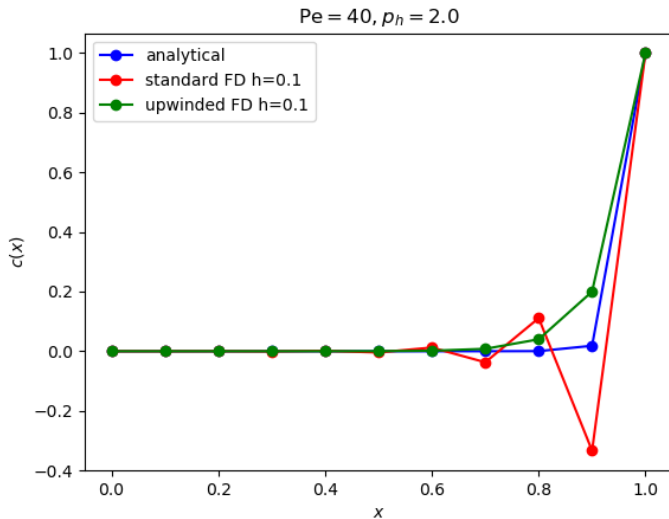
This FD equation has a monotonous solution for any $p_h > 0$. However, the approximation is $\mathcal{O}(h)$ now. Used mostly with shocks.

Linear algebraic problem

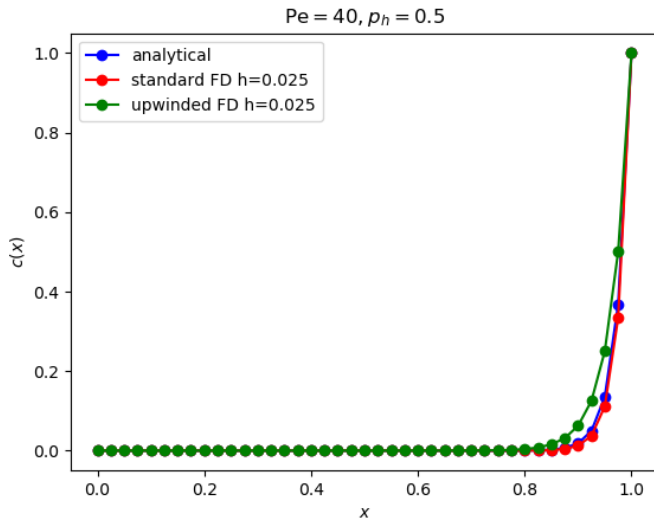
$$A\mathbf{c} = \mathbf{f}$$

$$\begin{bmatrix} 2+2p_h & -1 & 0 & 0 & \dots & 0 \\ -1-2p_h & 2+2p_h & -1 & 0 & \dots & 0 \\ \cdot & & & & & \cdot \\ \cdot & & & & & -1 \\ 0 & 0 & 0 & 0 & -1-2p_h & 2+2p_h \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ \cdot \\ \cdot \\ 1 \end{bmatrix}$$

Results with $h = 0.1$



Results with $h = 0.025$



Summary and questions

- Finite-Difference Method introduces a finite grid and approximates differential equations and boundary conditions at grid points with Finite-Difference formulas
- Application of FDM to linear BVP's, e.g., for Poisson equation, leads to a linear algebraic problem with the system matrix and right-hand side vector that depend on the BC's
- For some BVP's, e.g., convection-diffusion, lower-order FD approximations may lead to more stable numerical solutions
- QUESTION: perform an assembly of the FDM system for the BVP:

$$\begin{aligned}-ku'' &= f, \quad x \in (a, b) \\ -ku'(a) &= \mu_a \\ -ku'(b) + \beta u(b) &= \mu_b\end{aligned}$$

where $k > 0$ and $\beta > 0$ are constants.